

5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t$, $t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t$, $t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

$$\begin{aligned} \text{a. } \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum t)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T} \\ &= \frac{T(T+1)}{2} \left(\frac{2T+1}{3} - \frac{T+1}{2} \right) \\ &= \frac{T(T+1)}{2} \left(\frac{4T+2-3T-3}{6} \right) \\ &= \frac{T(T+1)(T-1)}{12} \\ &= \frac{T(T^2-1)}{12} \end{aligned}$$

$$\begin{aligned} \text{Var}(b_2|\mathbf{x}) &= \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)} \end{aligned}$$

$$\therefore \lim_{T \rightarrow \infty} \text{Var}(b_2|\mathbf{x}) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T^3 - T} = 0$$

$$\begin{aligned} \text{b. } x_t &= 0.5^t \\ \sum (x_t - \bar{x})^2 &= \sum x_t^2 - \frac{(\sum x_t)^2}{T} \\ \text{① } \sum x_t^2 &= \sum (0.5^t)^2 = \sum_{t=1}^T 0.25^t \\ &= \frac{0.25(1 - 0.25^T)}{1 - 0.25} = \frac{0.25}{0.75} (1 - 0.25^T) \\ &= \frac{1}{3} (1 - 0.25^T) \end{aligned}$$

$$\begin{aligned} \text{② } \sum x_t &= \sum 0.5^t = \frac{0.5(1 - 0.5^T)}{1 - 0.5} \\ &= 1 - 0.5^T \Rightarrow \sum (x_t - \bar{x})^2 = \frac{1}{3} (1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T} \end{aligned}$$

$$\text{Var}(b_2|x) = \frac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}}$$

$$T \rightarrow \infty, \quad 0.25^T \rightarrow 0, \quad 0.5^T \rightarrow 0, \quad \frac{1}{T} \rightarrow 0$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|x) = \frac{\sigma^2}{\frac{1}{3}} = 3\sigma^2 \neq 0$$

C. (a) $X_t = t$

Since X_t increases over time, the variation in X diverges,

$$\text{so } \sum (X_i - \bar{X})^2 \rightarrow \infty$$

Thus $\text{Var}(b_2|x) \rightarrow 0$, and the estimator is consistent.

(b) $X_t = 0.5^t$

Since X_t converges rapidly to 0, the variation X is bounded, so

$$\sum (X_t - \bar{X})^2 \text{ does not diverge}$$

Thus $\text{Var}(b_2|x)$ does not go to 0, and the estimator is not consistent.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
- b. $\beta_1 + 2\beta_2 = 5$
- c. $\beta_1 - \beta_2 + \beta_3 = 4$

(a) $t\text{-statistic} = \frac{b_2 - \beta_2}{\text{se}(b_2)} = \frac{3}{2} = 1.5$

$$t_{60, 0.975} = 2$$

$$\therefore 1.5 < 2$$

$\therefore$ Do not reject H_0

(b) $t\text{-statistic} = \frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} = \frac{3}{\sqrt{11}} \approx 0.9045$

$$\begin{aligned} \text{Var}(b_1 + 2b_2) &= \text{Var}(b_1) + 4\text{Var}(b_2) + 4\text{cov}(b_1, b_2) \\ &= 3 + 4(4) + 4(-2) \\ &= 11 \end{aligned}$$

$\therefore 0.9045 < 2 \quad \therefore$ Do not reject H_0

$$\begin{aligned} \text{(c)} \quad t\text{-statistic} &= \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} \\ &= \frac{(2 - 3 - 1) - 4}{\sqrt{16}} \\ &= -1.5 \end{aligned}$$

$$\begin{aligned} \text{Var}(b_1 - b_2 + b_3) &= \text{Var}(b_1) + \text{Var}(b_2) + \text{Var}(b_3) \\ &\quad - 2\text{Cov}(b_1, b_2) - 2\text{Cov}(b_2, b_3) + 2\text{Cov}(b_1, b_3) \\ &= 3 + 4 + 3 - 2(-2) - 2(0) + 2(1) \\ &= 16 \end{aligned}$$

$$\therefore |-1.5| < 2$$

$\therefore$ Do not reject H_0

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2, \text{mean}} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2, \text{mean}}$ is unbiased.

$$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + e_1$$

$$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + e_2$$

$$\begin{aligned} \hat{\beta}_{2, \text{mean}} &= \frac{(\beta_1 + \beta_2 \bar{x}_2 + e_2) - (\beta_1 + \beta_2 \bar{x}_1 + e_1)}{\bar{x}_2 - \bar{x}_1} \\ &= \frac{\beta_2 (\bar{x}_2 - \bar{x}_1) + (e_2 - e_1)}{\bar{x}_2 - \bar{x}_1} \\ &= \beta_2 + \frac{e_2 - e_1}{\bar{x}_2 - \bar{x}_1} \end{aligned}$$

$$E(\hat{\beta}_{2, \text{mean}} | x) = \beta_2 + \frac{E(e_2 | x) - E(e_1 | x)}{\bar{x}_2 - \bar{x}_1}$$

According to MR2, $E(e_i | x) = 0$

$$\therefore E(\hat{\beta}_{2, \text{mean}} | x) = \beta_2$$

b. Derive an expression for $\text{var}(\hat{\beta}_{2, \text{mean}} | x)$.

$$\hat{\beta}_2 - \beta_2 = \frac{e_2 - e_1}{\bar{x}_2 - \bar{x}_1}$$

$$\begin{aligned} \text{Var}(\hat{\beta}_2 | x) &= \text{Var}\left(\frac{e_2 - e_1}{\bar{x}_2 - \bar{x}_1} \mid x\right) = \frac{\text{Var}(e_2 | x) + \text{Var}(e_1 | x) - 2\text{Cov}(e_1, e_2 | x)}{(\bar{x}_2 - \bar{x}_1)^2} \\ &= \frac{\frac{\sigma^2}{N} + \frac{\sigma^2}{N}}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \end{aligned}$$

$$\text{Var}(e_i) = \sigma^2, \quad \text{Cov}(e_i, e_j) = 0$$

$$\text{Var}(e_i|x) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N e_i\right) = \frac{1}{(N/2)^2} \sum_1^{N/2} \text{Var}(e_i) = \frac{N/2 \cdot \sigma^2}{N^2/4} = \frac{2\sigma^2}{N}$$

$$\text{Hence } \text{Var}(e_2|x) = \frac{2\sigma^2}{N}$$

- c. Write down an expression for $\text{var}(\hat{\beta}_{2, \text{mean}})$.

$$\text{Var}(\hat{\beta}_{2, \text{mean}}) = E[\text{Var}(\hat{\beta}_{2, \text{mean}}|X)] + \text{Var}(E[\hat{\beta}_{2, \text{mean}}|X])$$

$$E[\hat{\beta}_{2, \text{mean}}|X] \text{ is consistent, } \sigma^2 = 0$$

$$\text{Var}(\hat{\beta}_{2, \text{mean}}) = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$$

- d. Under what conditions will $\hat{\beta}_{2, \text{mean}}$ be a consistent estimator for β_2 ?

$$N \rightarrow \infty \quad \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \rightarrow 0$$

This requires $N(\bar{x}_2 - \bar{x}_1)^2$ go to infinity

$(\bar{x}_2 - \bar{x}_1)$ needs to converge to a strictly positive constant.

- f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2, \text{mean}}$ is consistent?

$$\text{Var}(b_2, \text{OLS}) < \text{Var}(\hat{\beta}_{2, \text{mean}})$$

This follows from the Gauss-Markov theorem,
since OLS is the most efficient linear unbiased estimator

As N increases, both variances decrease and approach 0,
implying that $\hat{\beta}_{2, \text{mean}}$ is consistent

- g. Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]

$$E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$$

$$\text{pf. } U(0, 10) \quad \sigma_x^2 = \frac{(10-0)^2}{12} = \frac{100}{12} \approx 8.333$$

According to LLN,

$$\frac{1}{s_x^2} \rightarrow \frac{1}{\sigma_x^2} = \frac{12}{100} = 0.12$$

$$E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \xrightarrow{p} 0.16$$

$$p.f.: \bar{x}_1 \xrightarrow{P} E[U(0,5)] = 2.5$$

$$\bar{x}_2 \xrightarrow{P} E[U(5,10)] = 7.5$$

$$(\bar{x}_2 - \bar{x}_1) \xrightarrow{P} 7.5 - 2.5 = 5$$

$$\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{P} \frac{4}{5^2} = \frac{4}{25} = 0.16$$

- h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

If the observations are not ordered, both halves are random samples from the same population

By LLN, $\bar{x}_1, \bar{x}_2 \xrightarrow{P} E(x)$, so $\bar{x}_2 - \bar{x}_1 \xrightarrow{P} 0$

Thus the denominator vanishes and the variance does not go to 0, implying the estimator is inconsistent